

MW-A-101 INSPECTION PLAN

1. First-article control plan

Lot: first tool trial, tooling change, cavity repair, resin change and process requalification.

Sample: one complete dimensional report per cavity plus five assembled sets unless engineering increases the sample.

Methods: calibrated CMM/vision system for profiles and coordinates; pin/plug gauges for bores; micrometer for walls; optical comparator for draft/radii.

Acceptance: drawing-specific limits control; reference dimensions are recorded but do not independently reject the lot.

Record actual values, equipment ID, calibration due date, operator, cavity, lot and disposition.

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2. Feature-id inspection inventory

-- MW-M-101 --

MW-M-101-D01 | DATUM SYSTEM | A = PCB support plane; B = left PCB locator; C = rear locator |
MODEL_XYZ; origin per design input | basic

MW-M-101-E01 | BREP ENVELOPE | 64.000 x 56.000 x 12.000 mm | min=(-32.000,-28.000,0.000) | reference

MW-M-101-M01 | MOLD PULL / PARTING | +Z / top rim maximum footprint | part-local | vendor DFM

MW-M-101-M02 | GENERAL DRAFT / RADII | 1.5 deg; Rext>=0.8; Rint>=0.5 | all unspecified molded
faces/edges | tooling tune

MW-M-101-M03 | SIDE ACTIONS | connector openings are conditional on the bound electronics interface
table | feature-specific | shutoff draft 3.0 deg target

MW-M-101-W01 | WALL / FLOOR | 2.00 / 2.00 | base section | 1.80 MIN

MW-M-101-I01 | PCB ENVELOPE / DATUM | 50.00 x 42.00 x 1.60 | PCB bottom Z=3.20 | +0.30 XY / +0.20 Z

MW-M-101-H01 | CASE PILOTS | DIA 2.15->1.95 | XYZ=[[-28.0, -24.0], [28.0, -24.0], [-28.0, 24.0]] |
strip-torque DOE

-- MW-M-102 --

MW-M-102-D01 | DATUM SYSTEM | A = top exterior plane; B = skirt center; C = poka-yoke corner |
MODEL_XYZ; origin per design input | basic

MW-M-102-E01 | BREP ENVELOPE | 64.400 x 56.400 x 8.000 mm | min=(-32.200,-28.200,0.000) | reference

MW-M-102-M01 | MOLD PULL / PARTING | -Z / lower rim maximum footprint | part-local | vendor DFM

MW-M-102-M02 | GENERAL DRAFT / RADII | 1.5 deg; Rext>=0.8; Rint>=0.5 | all unspecified molded
faces/edges | tooling tune

MW-M-102-M03 | SIDE ACTIONS | none until the electronics interface table is frozen | feature-specific
| shutoff draft 3.0 deg target

MW-M-102-W01 | LID WALL / HEIGHT | 2.00 / 8.00 | lid section | 1.80 MIN

MW-M-102-F01 | SKIRT FIT | clearance 0.15/side | perimeter | +0.10/-0.05

MW-M-102-H01 | 3X LID CLEARANCE / CBORE / TOWER POCKET | DIA 2.80; CBORE DIA 5.00; POCKET DIA 6.80 |
XYZ=[[-28.0, -24.0], [28.0, -24.0], [-28.0, 24.0]] | +0.10/-0

MW-M-102-RF01 | RF KEEP-OUT | polygon=[] | above PCB antenna | no metal/coating/vent

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3. Assembly verification

Verify BOM identity and quantity closure.

Verify each released transform against assembly datum features.

Run positive-volume interference analysis; no unexpected interference is permitted.

Confirm controlled openings are unobstructed and all keep-out rules are satisfied.

Retain drawing, native-reopen, interference and inspection evidence under the same revision hash set.